

# A process-based water balance model for semi-arid ecosystems: A case study of psammophytic ecosystems in Mu Us Sandland, Inner Mongolia, China



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## ABSTRACT

We developed a process-based water balance model for semi-arid ecosystems (PWBSA) driven by daily meteorological data. The actual canopy transpiration and soil evaporation processes were simulated separately based on the Penman–Monteith model, while considering the energy partition with combined leaf area index (LAI) and canopy coverage, process-based stomatal conductance, root distribution, and available soil water, among other parameters. A simple bucket soil water model that considered preferential flow was used to simulate the soil water content in four soil layers (0–10 cm, 10–20 cm, 20–40 cm, and 40–80 cm). As a case study, the model was applied to the Mu Us Sandland, a temperate semi-arid region of China, using parameters obtained from field experiments of two psammophytic communities. The model validation using the observed leaf transpiration and daily soil water content measured in 2012 and 2013 showed that the PWBSA model simulated the water balance well for both semi-arid ecosystems. Energy partition was sensitive to changes in the LAI and canopy coverage, implying that the model is suitable for application to semi-arid ecosystems with patchy canopies. Based on the water balance simulation of the two psammophytic ecosystems during 1955–2013, differences in annual evapotranspiration, transpiration, and deep soil water loss were observed between the fixed and semi-fixed psammophytic ecosystems due to variations in the LAI, canopy coverage, species ecophysiological traits, root distribution, and soil properties. Compared with other ecosystems in semi-arid regions, psammophytic ecosystems may be more sensitive to changes in rainfall patterns and should receive more attention in future global climate change research.

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## 1. Introduction

Arid and semi-arid ecosystems represent 42–56% of the world's land (Melillo et al., 1993; Schimel, 2010) and are important components of the global water cycle. The water balance in these water-limited ecosystems is sensitive to global climate change, including intra- and inter-annual fluctuations (Martínez-Vilalta et al., 2002; Jolly and Running, 2004). The biodiversity, ecological services, and stability of semi-arid ecosystems have received much attention due to water shortages and global climate change (Jolly and Running, 2004; Ágreda et al., 2015).

Large models have been developed to simulate the water balance (Kremer and Running, 1996; Eilers et al., 2007) and conduct water resource management research (She et al., 2014). Empirical models have been used widely at the field level (Rockström et al., 1998; Reynolds et al., 2000; She et al., 2014). However, as empirical approaches have been developed under a constrained range of environmental variables, extreme conditions such as high vapor pressure deficits, low water availability, and extreme temperatures challenge the ability of these models to accurately estimate water balances (Raab et al., 2015). To improve the model performance and explain the mechanisms driving the water cycle, process-based water balance models, most of which work at the ecosystem scale (Kremer and Running, 1996; Baird and Wilby, 1999), have become increasingly common. General ecosystem models, such as ForGro (Mohren, 1987), are usually highly complex; consider

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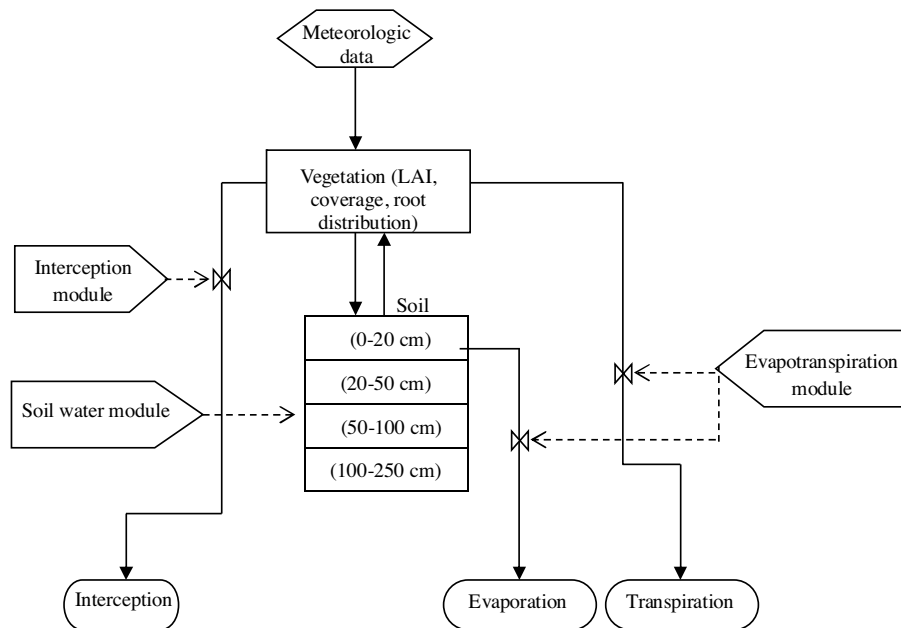


Fig. 1. Structure of the ecosystem-scale PWBSA model developed in this study.

water, carbon, and nitrogen cycles simultaneously; and contain many parameters and variables; thus, these models are difficult to apply (Tiktak and van Grinsven, 1995; Sawano et al., 2015). Several simple and widely applicable ecosystem models, such as FOREST-BGC (Running and Coughlan, 1988) and PnET (Aber and Federer, 1992), have since been developed to simulate plant growth and water balance separately, which connect these parameters via the soil water content. However, most of these models emphasize plant growth, and require too many parameters and variables to simulate the water cycle. To understand the water cycle on both ecosystem and landscape scales, specialized water balance models have been developed (Sawano et al., 2015) that calculate evapotranspiration and soil water movement with simplified models of plant growth based on daily meteorological data (Hall and Harding, 1993; Metcalfe and Buttle, 1999; Kite, 2000).

The water balance of semi-arid ecosystems differs distinctly from humid ecosystems (Kremer and Running, 1996); for example, evapotranspiration and soil water dynamics are given more attention in semi-arid ecosystems (Patanè, 2011; She et al., 2014; Sela et al., 2015). In addition, transpiration is often separated from evapotranspiration to evaluate water use efficiency (Eilers et al., 2007). Stomatal conductance is the main path through which plants control both leaf transpiration and carbon dioxide intake (Gao et al., 2002; Patanè, 2011; Raab et al., 2015), and root distribution can contribute greatly to water regulation through the soil–plant–air system (Martínez-Vilalta et al., 2002; Radersma and Ong, 2004; Eilers et al., 2007). These processes are often simulated in process-based water balance models (Kremer and Running, 1996; Jolly and Running, 2004). However, a number of simulation studies do not include any information on the mechanisms that explain the responses of the different plant species (Eilers et al., 2007). Sparse canopies and patchy patterns of semi-arid vegetation have a strong influence on latent heat flux partitioning (Kremer and Running, 1996; Raab et al., 2015). Moreover, soil water availability plays a major limiting role in arid and semi-arid regions (Jolly and Running, 2004; Eilers et al., 2007), while soil texture and structure may strongly regulate soil water movement (Hu and Si, 2014). Soil preferential flow is a main regulation mechanism in arid and semi-arid ecosystems with high proportions of macroporosity in the soil profiles (Granier et al., 1999; Li et al., 2009).

In this study, we developed a process-based water balance model based on the stomatal conductance behavior of dominant species to simulate the water balance of semi-arid ecosystems. A series of successional ecosystems, from shrub-dominated to grass-dominated communities, exists in Mu Us Sandland, China, and soil water availability is considered to be the key factor in the succession process (Guo, 2000; Li, 2012; Li et al., 2013). Therefore, we validated our process-based water balance model for semi-arid ecosystems (PWBSA) model with data collected from both shrub-dominated and grass-dominated ecosystems.

## 2. Materials and methods

### 2.1. Model framework and general assumptions

The PWBSA model can simulate canopy interception, soil water content, transpiration, and evaporation on a daily temporal scale (Fig. 1). Runoff is ignored, because there is almost no runoff in Sandland (Nian and Li, 2000). All of the variables and parameters used in the model are defined in Table 1. The meteorological data used as inputs include daily precipitation, mean temperature, maximum temperature, minimum temperature, mean relative humidity, and mean wind speed.

### 2.2. Interception module

Canopy interception can be calculated from the leaf area index (LAI) with the constant interception parameter  $I_{ck}$  ( $I_{ck}=0.25$ ) (Running and Coughlan, 1988):

$$I_c = I_{ck}LAI. \quad (1)$$

Variations in the LAI for the ecosystems are interpolated between the minimum and maximum LAI values. It is assumed that LAI increases linearly from zero to the maximum value during the leaf unfolding period, and the maximum value remains unchanged until the occurrence of leaf coloration, after which it decreases linearly to zero at the end of the growing season (Granier et al., 1999).

**Table 1**

Definition of the model variables and parameters.

| Symbol                           | Unit                                 | Definition                                                                       |
|----------------------------------|--------------------------------------|----------------------------------------------------------------------------------|
| <b>Interception module</b>       |                                      |                                                                                  |
| $I_c$                            | mm day <sup>-1</sup>                 | Canopy interception                                                              |
| $LAI$                            | m <sup>2</sup> m <sup>-2</sup>       | Leaf area index                                                                  |
| $I_{ck}$                         | /                                    | Coefficient of canopy interception                                               |
| <b>Soil water module</b>         |                                      |                                                                                  |
| $\Delta\theta_i$                 | v/v%                                 | Daily soil water recharge in $i^{th}$ layer                                      |
| $P$                              | mm                                   | Daily rainfall                                                                   |
| $ET_i$                           | mm                                   | Evapotranspiration from the soil in $i^{th}$ layer                               |
| $D_i$                            | mm                                   | Deeper infiltration from layer $i$ to layer $(i+1)$                              |
| $z_i$                            | m                                    | Soil depth of layer $i$                                                          |
| $K_{maxi}$                       | /                                    | Coefficient of preferential flow in $i^{th}$ layer                               |
| <b>Evapotranspiration module</b> |                                      |                                                                                  |
| $f_s$                            | /                                    | Partition coefficient of energy to soil                                          |
| $f_g$                            | /                                    | Partition coefficient of energy to the canopy                                    |
| $SCD$                            | /                                    | Canopy coverage                                                                  |
| $k$                              | /                                    | Light coefficient of extinction                                                  |
| $TR$                             | mm                                   | Canopy transpiration                                                             |
| $EV$                             | mm                                   | Soil evaporation                                                                 |
| $Rn$                             | J m <sup>-2</sup> s <sup>-1</sup>    | Net radiation                                                                    |
| $S$                              | J m <sup>-2</sup> d <sup>-1</sup>    | Soil heat flux                                                                   |
| $L$                              | J kg <sup>-1</sup>                   | Latent heat of vaporization                                                      |
| $\Delta$                         | kPa °C <sup>-1</sup>                 | Slope of the saturated vapor pressure curve at the temperature of the air        |
| $C_p$                            | J kg <sup>-1</sup> °C <sup>-1</sup>  | Specific heat of air at a constant pressure                                      |
| $\rho$                           | kg m <sup>-3</sup>                   | Air density                                                                      |
| $e_s$                            | kPa                                  | Saturated vapor pressure                                                         |
| $e$                              | kPa                                  | Vapor pressure                                                                   |
| $\gamma$                         | kPa °C <sup>-1</sup>                 | Psychrometric constant                                                           |
| $DAYL$                           | s d <sup>-1</sup>                    | Day length                                                                       |
| $ra$                             | m s <sup>-1</sup>                    | Corresponding aerodynamic resistance                                             |
| $rsc$                            | s m <sup>-1</sup>                    | Canopy surface resistance                                                        |
| $rss$                            | s m <sup>-1</sup>                    | Soil surface resistance                                                          |
| $g_i$                            | /                                    | Fraction of transpiration extracted from the soil of layer $i$                   |
| $h_i$                            | /                                    | Fraction of evaporation extracted from the soil of layer $i$                     |
| $r_i$                            | /                                    | Relatively extractable soil water in layer $i$                                   |
| $root_i$                         | /                                    | Root proportion in the soil of layer $i$                                         |
| $\theta_i$                       | v/v%                                 | Soil water content in layer $i$                                                  |
| $\theta_{fc}$                    | v/v%                                 | Soil field capacity                                                              |
| $\theta_w$                       | v/v%                                 | Soil wilting point                                                               |
| $z$                              | cm                                   | Soil layer depth                                                                 |
| $E$                              | cm                                   | Evaporation depth                                                                |
| $g_s$                            | mmol m <sup>-2</sup> s <sup>-1</sup> | Stomatal conductance                                                             |
| $g_v$                            | m s <sup>-1</sup>                    | Canopy conductance                                                               |
| $\psi_s$                         | kPa                                  | Soil water potential                                                             |
| $P_{ar}$                         | μmol m <sup>-2</sup> s <sup>-1</sup> | Photosynthetic active radiation                                                  |
| $d_{vp}$                         | /                                    | Relative vapor pressure deficit                                                  |
| $g_{om}$                         | mmol m <sup>-2</sup> s <sup>-1</sup> | Maximum possible stomatal conductance in the dark with zero soil water potential |
| $k_{\psi}$                       | /                                    | Elastic compliance of guard cell structure                                       |
| $k_{ab}$                         | mmol μmol <sup>-1</sup>              | Sensitivity of stomatal conductance to photosynthetic activity                   |
| $k_{\beta g}$                    | /                                    | Sensitivity of stomatal conductance to vapor pressure deficit                    |

### 2.3. Soil water module

Preferential flow occurs under the action of gravity if soil water is held in soil macropores (Martinez-Meza and Whitford, 1996; Granier et al., 1999; Li et al., 2009), is common in rocky or sandy soils (Li et al., 2009), and is a major feature of soil water dynamics (Wilcox et al., 2003; Li et al., 2009). Because the Richards equation is usually unsuitable for determining the soil water dynamics in sandy or rocky soil (Granier et al., 1999; Reynolds et al., 2000), we developed a simple bucket model combined with preferential flow to portray the sandy soil water dynamic in our model. The preferential flow assumes that rainfall can recharge the lower soil layers directly at a fixed proportion, regardless of whether the soil water is higher than the soil field capacity (Hornberger et al., 1985).

Four horizontal layers (0–10 cm, 10–20 cm, 20–40 cm, and 40–80 cm) were partitioned in the soil profile to represent the sand characteristics and root distribution. Horizontal heterogeneity in the soil was neglected. Deeper infiltration can occur when the soil water content in the upper layer is higher than the field capacity

of the soil (Eatherall, 1997; Cernusca et al., 1998). Daily soil water recharge in the first layer ( $\Delta q_1$ ) and the other three layers ( $\Delta\theta_i$ ) can be simulated with Eqs. (2) and (3), respectively:

$$\Delta\theta_1 = \frac{P(1 - \sum_{i=2}^4 K_{maxi} root_i) - ET_1 - D_1}{z_1}, \quad (2)$$

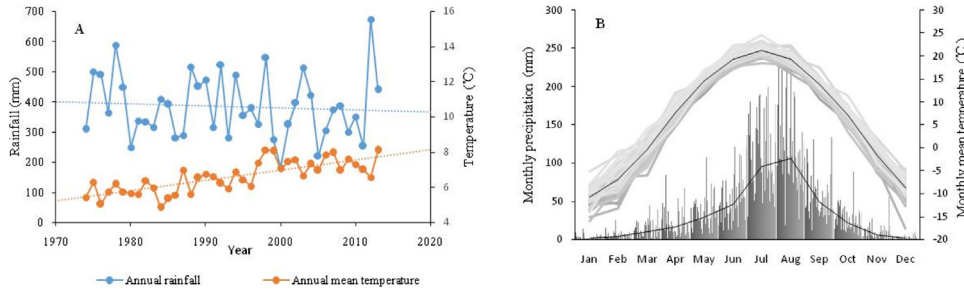
$$\Delta\theta_i = \frac{K_{maxi} root_i \cdot P + D_{i-1} - ET_i - D_i}{z_i}, \quad (3)$$

where  $D_1$  and  $D_i$  are calculated based on the soil field capacity, as shown below.

$$\text{If } P(1 - \sum_{i=2}^4 K_{maxi} root_i) - ET_1 \leq \theta_{fc1}, D_1 = 0.$$

$$\text{Else, } D_1 = P(1 - \sum_{i=2}^4 K_{maxi} root_i) - ET_1 - \theta_{fc1}.$$

$$\text{If } K_{maxi} root_i \cdot P + D_{i-1} - ET_i \leq \theta_{fci}, D_i = 0.$$



**Fig. 2.** Climate of Mu Us Sandy Land (1974–2013). (A) Mean annual rainfall and temperature. (B) Monthly mean temperature (light grey line) from 1974 to 2013 and the mean monthly temperature (bold line), as well as the monthly rainfall (histogram) from 1974 to 2013 and the monthly mean rainfall (bold line in histogram).

$$\text{Else, } D_i = K_{\max i} \text{root}_i \cdot P + D_{i-1} - ET_i - \theta_{fci}.$$

#### 2.4. Evapotranspiration module

Potential evapotranspiration was calculated using the Penman–Monteith equation. Evaporation and transpiration were separated by considering the LAI and canopy coverage. The process-based stomatal conductance model was scaled up to simulate canopy resistance combined with the LAI.

##### 2.4.1. Energy partition of evapotranspiration

LAI is normally used to derive the available energy to soil from the canopy (Nouvellon et al., 2000), although only a few models use canopy coverage (Wegehenkel, 2000). The former is suitable for homogenous communities with high LAI values, and the latter is better suited to those with low LAI values. However, problems may arise when these models are applied to arid/semi-arid ecosystems with patchy canopies (Raz-Yaseef et al., 2010), because some ecosystems have similar LAIs but different coverages, or have similar coverages but different LAIs. In these cases, the energy partition would differ between evaporation and transpiration. In this PWBSA model, the available energy below the canopy is calculated with the Beer–Lambert function (Granier et al., 1999). The partition coefficients of energy to the canopy or soil ( $f_s$  and  $f_g$  ( $f_s + f_g = 1$ ), respectively) are calculated based on the LAI and canopy coverage:

$$f_s = \frac{1 - SCD}{1 - 2SCD + SCD \cdot \exp(kLAI)}, \quad (4)$$

$$f_g = 1 - f_s, \quad (5)$$

where  $SCD$  is canopy coverage and  $k$  is the light coefficient of extinction ( $k = 0.5$ ).

Then, the canopy transpiration ( $TR$ ) and soil evaporation ( $EV$ ) can be calculated separately based on the Penman–Monteith equation:

$$TR = f_g \cdot \left( \frac{1}{L} \cdot \frac{\Delta(Rn - S) + C_p \rho (es - e)/ra}{\Delta + \gamma(1 + rsc/ra)} \cdot DAYL \right) \cdot \sum_{i=0}^n g_i, \quad (6)$$

$$EV = f_s \cdot \left( \frac{1}{L} \cdot \frac{\Delta(Rn - S) + C_p \rho (es - e)/ra}{\Delta + \gamma(1 + rss/ra)} \cdot DAYL \right) \cdot \sum_{i=0}^n h_i. \quad (7)$$

Actual evapotranspiration is regulated by soil water content and root vertical distribution. As such,  $g_i$  and  $h_i$  are calculated as follows (Rockström et al., 1998; Wegehenkel, 2000):

$$g_i = r_i \text{root}_i, \quad (8)$$

$$h_i = r_i h'_i, \quad (9)$$

where

$$r_i = \frac{\theta_i - \theta_w}{\theta_{fc} - \theta_w}, \quad (10)$$

$$\int_0^E h'(z) dz = 1, \text{ and } h'(z) = h_0 \frac{1 - z/E}{1 + 20z/E}, \quad (11)$$

Soil water evaporation is affected by meteorological factors and soil water content. The maximum evaporation depth of sandy soil is about 10 cm in the study region (Liu et al., 2006; Wang et al., 2006). In the model, the depth of evaporation was set to 10 cm.

##### 2.4.2. Process-based canopy conductance module

A process-based stomatal conductance ( $g_s$ ) model (Gao et al., 2002) was applied to simulate gas exchange per leaf area, as given below:

$$g_s = \frac{g_{om} + k_\psi \psi_s + k_{\alpha\beta} P_{ar}}{1 + k_{\beta g} d_{vp}} \quad (12)$$

A canopy conductance model (Olioso et al., 1996) was then applied to scale up from the leaf level to the ecosystem level by linking the LAI and stomatal conductance model. Next, the canopy surface resistance ( $rsc$ ) is calculated as the reciprocal of canopy conductance ( $g_v$ ):

$$g_v = g_s LAI / (0.5LAI + 1), \text{ then, } rsc = 1/g_v \quad (13)$$

#### 2.5. Study site and data description

Mu Us Sandland has an area of about 40,000 km<sup>2</sup>, is located in northwestern China (N37°30'–N39°20'; E107°20'–E111°30'), and has a temperate semi-arid climate. The annual mean temperature is 6–8.5 °C (1974–2013), and the annual mean precipitation is 372.9 mm (range: 177.4–678.7 mm), with 77% of precipitation occurring from June to September as rain (Fig. 2). Soil water content is thought to be a main driver of psammophytic succession (Guo, 2000; Li, 2012; Li et al., 2013). There are five successional stages of psammophytic ecosystems in Mu Us Sandland (Guo, 2000). The two communities selected for our case study to simulate water balance: (1) semi-fixed dunes with *Artemisia ordosica* (CSF) and (2) fixed dunes with *A. ordosica* + *Stipa bungeana* (CF), are distributed extensively (Wang et al., 1997; Guo, 2000). *A. ordosica* is a deciduous sub-shrub in the Asteraceae family with a deep taproot and slightly fleshy, cleft linear leaf, while *S. bungeana* is a dominant grass in the Gramineae family of the temperate steppe with fibrous roots and narrow needle-like leaves.

The field experiment was conducted at the Sand Land Grassland Ecological Research Station in Ordos Plateau, Institute of Botany, Chinese Academy of Sciences, China (N39°29.66', E110°11.47'). General features of the two communities were measured during the growing season (Table 2). Fine roots (<2 mm) in the four soil layers (0–10 cm, 10–20 cm, 20–40 cm, and 40–80 cm) were dug up and sieved. The gas exchange process (LI-6400; LI-COR Biosciences, Inc., Lincoln, NE, USA) and leaf water potential (WP4; Decagon, Pullman, WA, USA) were measured during the growing season in 2012 and 2013.

**Table 2**

General features of the two studied communities in Mu Us.

| No. | Type                                                  | Coverage (%) | Height (m) | $LAI_{max}$ ( $m^2 m^{-2}$ ) | Species richness |
|-----|-------------------------------------------------------|--------------|------------|------------------------------|------------------|
| CSF | <i>A. ordosica</i> in Semi-fixed dune                 | 43           | 0.75       | 0.75                         | 5                |
| CF  | <i>A. ordosica</i> + <i>S. bungeana</i> in fixed dune | 65           | 0.41       | 1.08                         | 21               |

**Table 3**

Soil properties, root distribution, and species stomatal parameters of the two communities.

| Parameters                | CSF     | CF      |
|---------------------------|---------|---------|
| $\theta_{fc1}$ (0–10 cm)  | 9.79    | 16.5    |
| $\theta_{fc2}$ (10–20 cm) | 12.59   | 16.5    |
| $\theta_{fc3}$ (20–40 cm) | 9.21    | 16.5    |
| $\theta_{fc4}$ (40–80 cm) | 5.7     | 9.8     |
| $\theta_w$                | 0.3     | 1.4     |
| $Root_1$ (0–10 cm)        | 0.38    | 0.38    |
| $Root_2$ (10–20 cm)       | 0.11    | 0.06    |
| $Root_3$ (20–40 cm)       | 0.31    | 0.26    |
| $Root_4$ (40–80 cm)       | 0.20    | 0.30    |
| $g_{om}$                  | 501.43  | 1231.03 |
| $k_{\psi}$                | 0.05    | 5.23    |
| $k_{\alpha\beta}$         | 3.12    | 0.61    |
| $k_{pg}$                  | 1019.23 | 214.28  |

In both communities, about 44–49% of the fine roots were found in the upper 20 cm, while 31% of fine roots in CSF were in the 20–40 cm layer, and 30% of fine roots of CF were in the 40–80 cm (Table 3). Soil water is simulated in the four soil layers in the model (0–10 cm, 10–20 cm, 20–40 cm, and 40–80 cm). Therefore, soil texture, bulk density, and pressure–volume curves were measured in each layer (Table 3). The soil water content was measured automatically at 10-min intervals (ECH20 EC-5; Decagon, Pullman, WA, USA) from June 2011 to December 2013 in each layer. The soil water

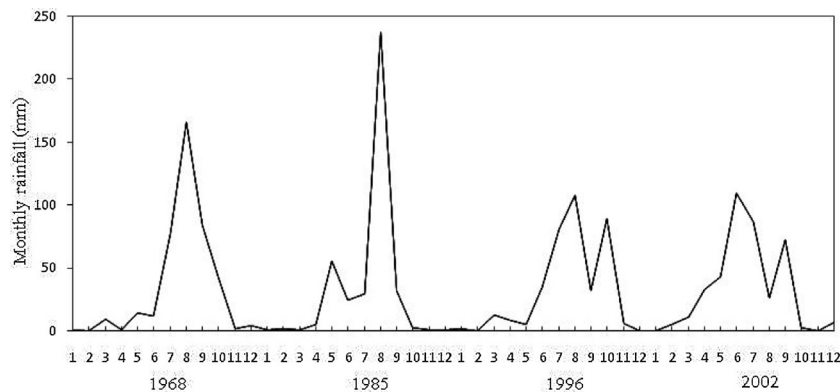
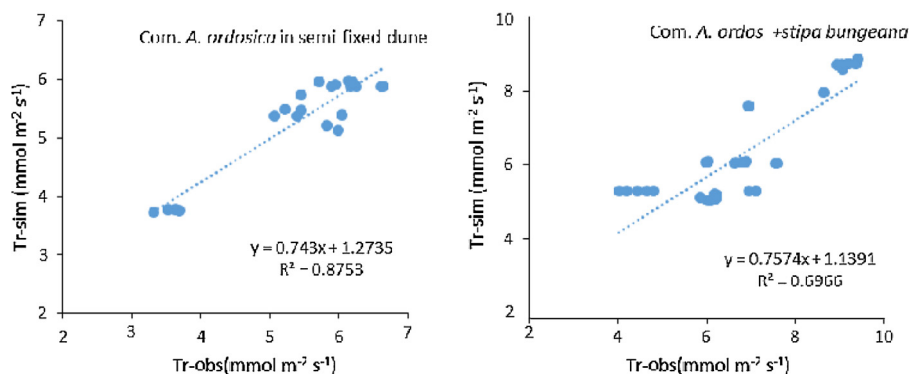
content data were calibrated according to the empirical calibration equation to reach approximately  $\pm 1$ –2% accuracy for specific soil. Groundwater replenishment was ignored in the model because the sand sediment is very thick (about 10–30 m in depth) (Peking University et al., 1983). Sandy soils have porosities of about 70% (Peking University et al., 1983).

Daily meteorological data over 58 years (1955–2013) were obtained from the China Meteorological Administration, including precipitation, mean, maximum, and minimum air temperature, mean relative humidity, and wind speed. Daily horizontal radiation was calculated with the model of Thornton and Running (1999). Then, photosynthetic active radiation was calculated with an empirical equation (Zhou et al., 1996). Among the 40 years, we selected four typical years with similar annual rainfalls and two rainfall patterns to simulate the water balance of the psammophytic ecosystems (Fig. 3).

### 3. Results and discussion

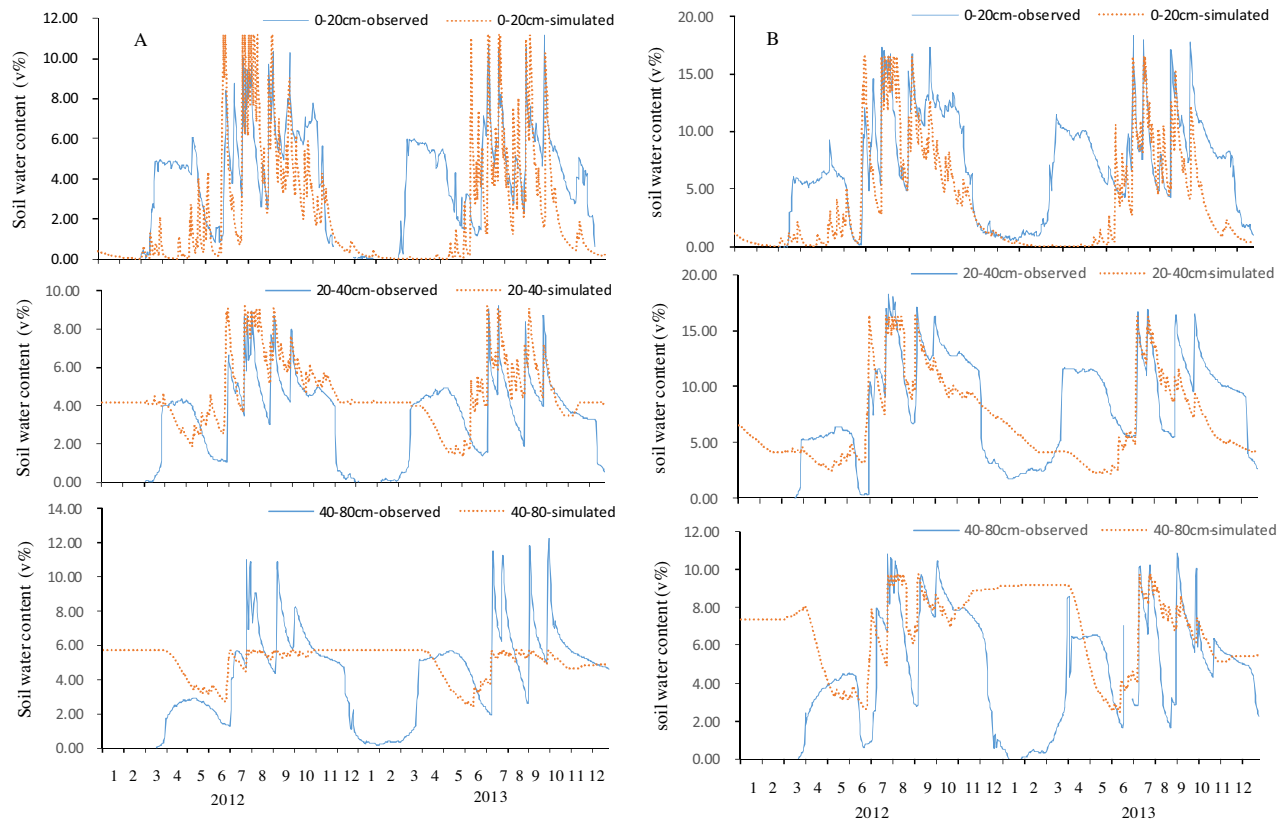
#### 3.1. Model validation and sensitivity analysis

The parameters of the process-based stomatal conductance model (Gao et al., 2002) were validated with field measurements. The hourly transpiration rate of the two communities was well simulated (Fig. 4). Stomatal conductance is commonly simulated to calculate canopy conductance, transpiration, and photosynthesis in

**Fig. 3.** Four typical rainfall years with two different rainfall patterns used to simulate the psammophytic ecosystems.**Fig. 4.** Leaf transpiration rate observed and simulated with the process-based stomatal model.

**Table 4**  
Comparison of the measured transpiration (T) and simulated transpiration of the two psammophytic ecosystems examined in this study.

| Vegetation                              | Measured T<br>(mmol m <sup>-2</sup> s <sup>-1</sup> ) | Simulated T<br>(mmol m <sup>-2</sup> s <sup>-1</sup> ) | Error                                   |      |
|-----------------------------------------|-------------------------------------------------------|--------------------------------------------------------|-----------------------------------------|------|
|                                         |                                                       |                                                        | (mmol m <sup>-2</sup> s <sup>-1</sup> ) | %    |
| <i>A. ordosica</i>                      | 5.65                                                  | 5.38                                                   | −0.27                                   | −4.7 |
| <i>A. ordosica</i> + <i>S. bungeana</i> | 6.82                                                  | 6.31                                                   | −0.52                                   | −7.6 |



**Fig. 5.** Observed and simulated daily soil water contents of three soil layers in the two communities in 2012 and 2013: (A) *A. ordosica* in semi-fixed dunes and (B) *A. ordosica* + *S. bungeana* in fixed dunes.

land surface models (Jolly and Running, 2004) and water balance models (Rockström et al., 1998). Empirical stomatal conductance models are more popular for scaling up from the leaf to ecosystem and even watershed levels, due to its relatively simple and linear relationship with environmental drivers (Rockström et al., 1998). However, process-based models are increasingly common in ecosystem and global models (Sitch et al., 2003; Wang et al., 2016). The stomatal model by Gao et al. (2002) is based on the ecophysiological mechanism of the stomata, which can be used to portray different stomatal dynamics among species. *A. ordosica*, found in both communities in this study, had different ecophysiological behaviors depending on the ecological features (Table 2) and environmental factors (Table 3) (Wang et al., 1997; Zhang et al., 2007), and thus could be simulated with the stomatal model (Fig. 4, Table 4).

The simulated and observed daily soil water contents of the two communities were compared in three soil layers (0–20 cm, 20–40 cm, and 40–80 cm) in 2012 and 2013 (Fig. 5, Table 5). In both communities, the soil water content was simulated well in the growing season, while it was higher or lower than the observed soil water content in the non-growing season (Table 5). The simulated soil water content was lower than the observed soil water content in May (Fig. 5), possibly due to seasonal freezing during the non-growing season in the study area (Han et al., 2015), which

was not accounted for in the PWBSA model. Although the annual and growing season soil water contents were not affected greatly, it would be better to consider soil freezing and melting processes in the future.

The combined LAI and canopy coverage energy partition formula is used in the PWBSA model to consider ecosystems with patchy canopies. We found that canopy transpiration and evapotranspiration were sensitive to the LAI and canopy coverage, respectively (Fig. 6). Canopy transpiration and evapotranspiration increased distinctly with increasing canopy coverage under the same LAI or with increasing LAI under the same canopy coverage. Patchy canopies are a common feature of arid/semi-arid ecosystems (Reynolds et al., 2000; Eilers et al., 2007), which experience evapotranspiration that changes with different LAIs or canopy coverage (Reynolds et al., 2000; Raz-Yaseef et al., 2010). The energy partition formula could help portray evapotranspiration in arid/semi-arid ecosystems.

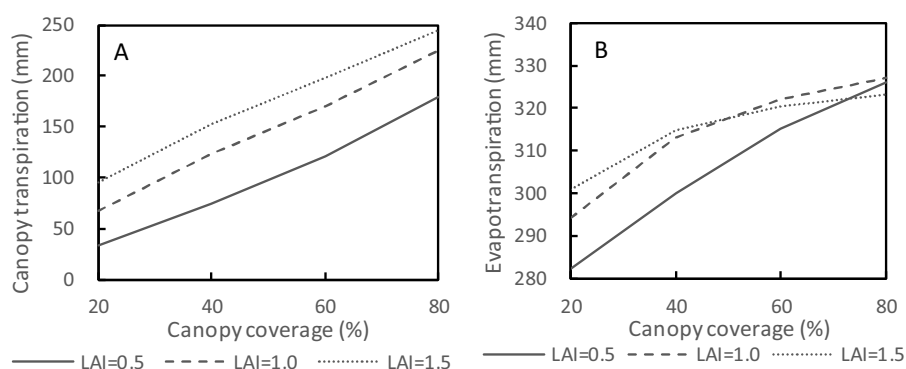
3.2. Water balance of two psammophytic ecosystems at different successional stages

The annual mean water balance features of the two communities were analyzed based on a daily water balance simulation from 1955 to 2013 (Table 6). The annual mean and maximum actual evapotranspiration values of CF were higher than those of CSF.

**Table 5**

Comparison of measured and simulated soil water content (SW) of two psammophytic ecosystems.

| Vegetation                              | Period                     | Soil depth (cm) | Measured SW | Simulated SW | Error  |       |
|-----------------------------------------|----------------------------|-----------------|-------------|--------------|--------|-------|
|                                         |                            |                 | (%v/v)      | (%v/v)       | (%v/v) | %     |
| <i>A. ordosica</i>                      | Jun–Oct 2012–2013          | 0–20            | 5.23        | 4.54         | –0.68  | –13.1 |
|                                         |                            | 20–40           | 4.39        | 5.74         | +1.35  | +30.9 |
|                                         |                            | 40–80           | 5.75        | 5.01         | –0.74  | –12.8 |
|                                         | Jan–May, Nov–Dec 2012–2013 | 0–20            | 3.91        | 0.68         | –3.23  | –82.6 |
|                                         |                            | 20–40           | 3.68        | 3.37         | –0.29  | –8.0  |
|                                         |                            | 40–80           | 4.05        | 4.58         | +0.53  | +13.1 |
| <i>A. ordosica</i> + <i>S. bungeana</i> | Jun–Oct 2012–2013          | 0–20            | 9.48        | 10.54        | +1.06  | +11.2 |
|                                         |                            | 20–40           | 6.13        | 7.55         | +1.43  | +23.3 |
|                                         |                            | 40–80           | 9.77        | 6.88         | –2.88  | –29.5 |
|                                         | Jan–May, Nov–Dec 2012–2013 | 0–20            | 5.13        | 6.75         | +1.63  | +31.8 |
|                                         |                            | 20–40           | 3.97        | 0.93         | –3.04  | –76.5 |
|                                         |                            | 40–80           | 4.82        | 6.89         | +2.06  | +42.7 |

**Fig. 6.** Sensitivity of canopy transpiration and evapotranspiration to changes in the LAI and canopy coverage. (A) Simulated canopy transpiration under different LAIs and canopy coverages. (B) Simulated evapotranspiration under different LAIs and canopy coverages.**Table 6**Annual mean simulated water balance of *A. ordosica* in semi-fixed dune and *A. ordosica* + *S. bungeana* in fixed dune. Mean annual precipitation, MAP; actual evapotranspiration, AET; transpiration, T; evaporation, E.

|                                                        | MAP (mm) | AET (mm) | AET/MAP (%) | Deep loss (mm) | T (mm) | T/AET (%) | E (mm) | E/AET (%) |
|--------------------------------------------------------|----------|----------|-------------|----------------|--------|-----------|--------|-----------|
| <i>A. ordosica</i> in semi-fixed dunes                 |          |          |             |                |        |           |        |           |
| Mean                                                   | 384.16   | 322.85   | 84          | 61.31          | 112.95 | 35        | 209.90 | 65        |
| Minimum                                                | 181.52   | 183.23   | 101         | –7.88          | 69.95  | 38        | 113.28 | 62        |
| Maximum                                                | 673.26   | 440.13   | 65          | 237.37         | 144.79 | 32        | 305.18 | 68        |
| <i>A. ordosica</i> + <i>S. bungeana</i> in fixed dunes |          |          |             |                |        |           |        |           |
| Mean                                                   | 384.16   | 355.10   | 92          | 29.05          | 194.86 | 55        | 160.24 | 45        |
| Minimum                                                | 181.52   | 179.72   | 99          | –14.79         | 94.12  | 52        | 85.60  | 48        |
| Maximum                                                | 673.26   | 532.57   | 79          | 142.63         | 291.24 | 55        | 241.33 | 45        |

Moreover, the ratios of the mean or maximum actual evapotranspiration to annual mean precipitation of CF were higher than those of CSF. In addition, the ratios of the mean or maximum canopy transpiration to actual evapotranspiration were higher than those of CSF. The differences between the two ecosystems may have been caused by the higher LAI and canopy coverage in CF than in CSF (Table 2), leading to more soil water drawn out by evapotranspiration, and more deeper soil water extracted for eventual evapotranspiration due to the deeper root distribution (Table 3), leading to less deep soil water loss in CF.

Deep soil water loss is predominantly affected by the water holding capacity and root depth (Eilers et al., 2007). The amount of available water is a main driver for vegetation succession in psammophytic ecosystems in Mu Us Sandland and other semi-arid regions (Reynolds et al., 2000; Khaertidova and Longobardi, 2013; Ágreda et al., 2015), as interactions between soil water storage and vegetation promote community succession (Zavala et al., 2000; Reynolds et al., 2000). In addition, changes in the LAI, canopy coverage, species ecophysiological traits, root distribution,

and soil properties may result in variations in water partitioning among transpiration, evaporation, soil water vertical distribution, and deep loss in the ecosystem water balance (Rockström et al., 1998; Bellot et al., 1999; Reynolds et al., 2000). In the future, plant growth and species competition should be considered in the model to better understand the mechanisms of vegetation succession.

### 3.3. Evapotranspiration of psammophytic ecosystems under different rainfall patterns

Because global climate change models have predicted that temperature will continue to increase and rainfall patterns will change (Raab et al., 2015), ecosystem responses to rainfall patterns are drawing increasing attention (Corradini et al., 1997; Reynolds et al., 2000; Fay et al., 2002; Raz-Yaseef et al., 2012). In this study, we selected four years (1968, 1985, 1996, and 2002) with similar annual rainfalls (377.9–396.9 mm) to analyze the response of evapotranspiration in psammophytic ecosystems to rainfall patterns. There was higher rainfall in autumn (August–October) in 1968 and

**Table 7**Simulated water balance of *A. ordosica* in four years with typical rainfall. Mean annual precipitation, MAP; actual evapotranspiration, AET; transpiration, T.

| Year | MAP (mm) |         |         | AET (mm) | T (mm) | Deep soil water loss (mm) | AET/MAP (%) | T/AET (%) |
|------|----------|---------|---------|----------|--------|---------------------------|-------------|-----------|
|      | Annual   | Apr–Jul | Aug–Oct |          |        |                           |             |           |
| 1968 | 395.5    | 103.1   | 290.5   | 278.2    | 121.1  | 117.3                     | 70.3        | 43.5      |
| 1985 | 391.2    | 114.7   | 271.6   | 256.2    | 110.0  | 135.0                     | 65.5        | 42.9      |
| 1996 | 377.9    | 128.8   | 228.8   | 300.5    | 110.4  | 77.4                      | 79.5        | 36.7      |
| 2002 | 396.9    | 272.8   | 100.8   | 348.4    | 132.5  | 48.5                      | 87.8        | 38.0      |

1985 (autumn rain year), and higher rainfall in spring (April–July) in 1996 and 2002 (spring rain year) (Fig. 3, Table 6). The annual mean evapotranspiration in the spring rain years was higher than that in the autumn rain years (Table 7). The proportion of annual evapotranspiration to rainfall was 79.5–87.8% in the spring rain years, and 65.5–70.3% in the autumn rain years. However, the proportion of annual evapotranspiration to rainfall was higher in an autumn rain year in a rain-fed maize ecosystem (Hao et al., 2013). Therefore, this parameter should be determined based on plant phenology and functional type (Reynolds et al., 2000; Fay et al., 2002; Raz-Yaseef et al., 2012). More soil water recharged into deeper soil layers in autumn rain years. Given that more intra-annual rainfall evaporation occurred in spring rain years, the net primary productivity may increase in semi-arid/arid regions (Lu et al., 2016). Increases in the normalized difference vegetation index and vegetation coverage have been observed during past climate change in arid/semi-arid regions (Herrmann et al., 2005; Fensholt et al., 2012). Meanwhile, the dominant species of semi-arid/arid ecosystems is likely to change in response to changes in long-term rainfall patterns (Reynolds et al., 2000). Actual evapotranspiration may vary more drastically in response to changing rainfall patterns in psammophytic ecosystems than other ecosystems due to their particular soil properties (Laio et al., 2001; Pan and Peters-Lidard, 2008). Finally, ecosystems in semi-arid regions may become more fragile due to a reduction in the inter-annual adjusting capacity of soil water in response to increased rainfall evaporated intra-annually with more frequent spring rainfalls (Ágreda et al., 2015). Therefore, future studies should examine the response of psammophytic ecosystems in semi-arid regions to changes in climate.

#### 4. Conclusions

In this study, we developed a PWBSA model driven by daily meteorological data. To address the patchy canopies of arid/semi-arid ecosystems, we derived a new energy partitioning formula by combining the effects of the LAI and canopy coverage and simulated transpiration and evaporation separately in the model. As a case study, the model was applied to two psammophytic ecosystems, CF and CSF, in Mu Us Sandland, China. The model showed good performance, and may be applied to arid and semi-arid regions for ecosystem- and landscape-scale studies. The results showed clear differences in annual mean and maximum transpiration, actual evapotranspiration, and deep soil water loss between the two studied psammophytic communities. CF used more soil water for transpiration, and had less deep soil water loss than CSF. The simulated response of actual evapotranspiration to rainfall patterns indicated that the psammophytic ecosystems were highly sensitive to changes in rainfall pattern. Future research should focus on the responses of psammophytic ecosystems to global climate change.

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